

PA 2 - Shakespeare Text Generation Using N-grams

Due May 21 11:59 PM

Overview

In this assignment you will implement n-gram models and train them to generate text using Shakespeare's past works. All graded tasks are marked with TODO comments in `PA2_starter.ipynb`.

Learning Goals

- Understand text preprocessing and tokenization for language modeling.
- Implement a Unigram model (frequency-based word generation).
- Implement a Bigram model (next-word prediction with one word of context).
- Generalize to an N-gram model with Laplace smoothing.
- Evaluate language models using perplexity.
- Generate Shakespeare-like text from your trained models.

Allowed Libraries

- Only the already imported packages are allowed.

Deliverables

- Submit `PA2_starter.ipynb` with all TODOs completed.
- Do not rename the notebook (the autograder expects this filename).

Point Distribution (100 points)

Part	Points
Part 1: Text Preprocessing and Tokenization	10
Part 2: Unigram Language Model	40
Part 3: Bigram Language Model	50
Part 4 (Extra Credit): N-gram Language Model	20

Notes

- Some functions are already provided.

Submission Checklist

- All TODOs are implemented.
- Notebook runs end-to-end without errors.
- You did not change function signatures.

Tips

- Submit each part as you go, as previous functions are used in later parts of the assignment.